



Croatian Open Competition in Informatics

Round 3, January 14th 2023

Tasks

Task	Time limit	Memory limit	Score
Estimathon	1 second	512 MiB	50
Dirigent	1 second	512 MiB	70
Baltazar	4 seconds	512 MiB	110
Bomboni	2 seconds	512 MiB	110
Skrivača	2 seconds	512 MiB	110
Total			450



Task Estimathon

Paula and Domagoj are organizing an exciting team competition called Estimathon. They have already prepared the questions, and now they are arranging the room for the quiz.

Before you could blink, Domagoj had already placed n tables in the room. Now they need to add chairs. They have very colourful chairs. They come in m colours and there are a_i chairs of the i -th colour. Every team consists of **four** people. Therefore, for each table, Paula and Domagoj need to add four chairs. They want to make the room look as nice as possible. To make that happen, they agreed on these conditions:



- Each table has all chairs of the same colour
- Every chair colour is used, that is, for every colour there exists at least one table that has chairs of that colour

Caring too much about how nice the room will look, they suddenly realized that the quiz starts very soon! Help them find out if a chair arrangement with these conditions is even possible.

Input

The first line contains integers n and m ($1 \leq n, m \leq 100$), the number of tables and the number of chair colours.

The second line contains m integers a_i ($1 \leq a_i \leq 100$), where the i -th number is the number of chairs of the i -th colour.

Output

In the first and only line output DA if it is possible to arrange the chairs so that both conditions hold, otherwise output NE.

Scoring

Subtask	Points	Constraints
1	11	$a_1 = \dots = a_m = 4$
2	39	No additional constraints.

Examples

input

7 3
5 21 9

output

DA

input

5 4
8 5 10 3

output

NE

input

6 5
5 5 5 5 5

output

NE

Clarification of the second example:

We can have 5 tables with chairs of the same colour, but we cannot have a table with colour 4, so the second condition can't be satisfied.



Task Dirigent

Winter school of informatics ends with a traditional dance. There are n students who participate. Each of them has a unique label between 1 and n .

First, conductor Krešo orders the students to form a circle such that each student holds hands with two other students.

Alenka is wondering if it is possible to break the circle by making exactly one pair of neighbouring students stop holding hands and that the newly formed sequence of students is sorted by their labels. For example, if their order is 3 4 1 2, than the circle can be broken between students with labels 4 and 1, but if their order is 2 1 4 3, than there is no way to break the circle in such way.



During the night Krešo is going to give q instructions. In each of them, he is going to order two students to swap places. After each swap you need to help Alenka answer her question.

Input

The first line contains two integers n and q ($1 \leq n, q \leq 300\,000$), the number of students and the number of swaps.

The second line contains n integers a_i ($1 \leq a_i \leq n$), describing the initial placement of students in the circle.

In each of the next q lines there are two integers x_i, y_i ($1 \leq x_i, y_i \leq n, x_i \neq y_i$), that describe Krešo's i -th instruction in which students with labels x_i and y_i swap places.

Output

In the i -th of the q lines output the answer to Alenka's question after i swaps have been carried out. If the answer is affirmative output **DA**, otherwise **NE**.

Scoring

Subtask	Points	Constraints
1	15	$n, q \leq 500$
2	20	$n, q \leq 5\,000$
3	35	No additional constraints.



Examples

input

5 2
2 3 4 5 1
1 3
3 1

output

NE
DA

input

4 2
2 3 1 4
4 2
3 4

output

NE
DA

input

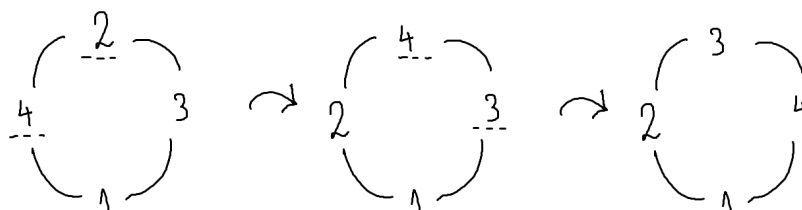
6 5
2 1 5 6 3 4
3 1
3 4
3 2
4 5
5 4

output

NE
NE
DA
NE
DA

Clarification of the second example:

Students in the beginning, after the first and after the second swap.





Task Baltazar

Baltazar decided to go on a vacation. Currently, he is in Baltazargrad, and wants to travel to Primošten. To get there, he has to go through many cities. There are n cities, and they are connected with m two-way roads. Baltazargrad is labeled as city no. 1, and Primošten as city no. n .

Baltazar isn't sure about the route from Baltazargrad to Primošten, so he will use GPS. It will lead him to his destination using the **shortest** route.

But Baltazar really likes to travel, and he can pour his magic potion on any road (even the ones he won't pass by), and increase its length by **2** kilometers. He can pour it on **only one** road.

Soon he realized that he has to check-in in the hotel Zora in Primošten before noon, so he can't increase the length of the shortest route too much. Now, he wants to know how many roads can he pour his magic potion on, so that the shortest distance between Baltazargrad and Primošten increases by **exactly 1** kilometer.

Help him determine the roads he can pour his magic potion on.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \leq t \leq 10\,000$). The description of the test cases follows.

The first line of each test case contains integers n and m ($2 \leq n \leq 300\,000$, $1 \leq m \leq \min(300\,000, \frac{n \cdot (n-1)}{2})$), the number of cities, and the number of roads between cities.

The following m lines contain integers a_i , b_i and w_i ($1 \leq a_i, b_i \leq n$, $a_i \neq b_i$, $1 \leq w_i \leq 10^9$), meaning there is a road between cities a_i and b_i , and its length is w_i . Between each pair of cities, there is at most one road connecting them.

All the cities are connected, i.e., for each pair of cities, there is a path from one to another, but not necessarily direct.

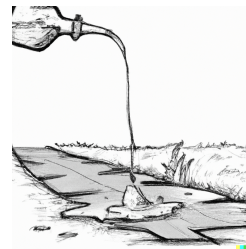
It is guaranteed that the sum of n over all test cases does not exceed 300 000, and that the sum of m over all test cases does not exceed 300 000.

Output

In the first line, print the integer c , the number of roads on which Baltazar can pour his magic potion. In the second line, print c integers, the indices of the roads in **increasing** order.

Scoring

Subtask	Points	Constraints
1	15	$n, m \leq 1\,000$
2	30	There is a road between Baltazargrad and Primošten, and its length is 1 kilometer longer than the shortest distance between these cities.
3	65	No additional constraints.





Examples

input

```
3
6 6
1 2 2
1 3 2
2 4 2
3 5 2
4 5 1
5 6 2
6 6
1 2 2
1 3 2
2 4 2
3 5 2
4 5 3
5 6 2
6 7
1 2 2
1 3 2
2 4 2
3 5 2
4 5 1
5 6 2
1 6 7
```

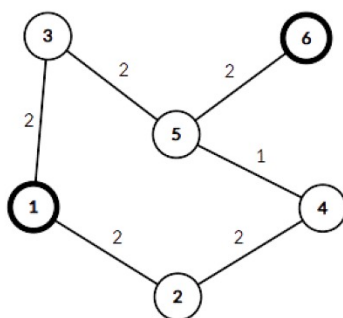
output

```
2
2 4
0

3
2 4 6
```

Clarification of the example:

The cities and the roads are shown in the image. If Baltazar pours his magic potion on road 2 (between cities 1 and 3), or on road 4 (between cities 3 and 5), then the shortest distance between cities 1 and n will increase by 1.





Task Bomboni

Iva is a big fan of candy! In front of her is an n times n field filled with candy and obstacles. Iva is currently in the upper left cell of the field and by moving only down and right she will travel to the lower right cell. The cell Iva is currently in does not contain an obstacle.



In every cell, there is either an obstacle or a piece of candy with a number written on it. Iva will eat all the candy she gets her hands on during her trip (including the candy in the first and last cell) and then multiply all the numbers on them. Iva knows her favourite number is k and she wants the product of the numbers on the candy she has eaten to be divisible by k . She wants to know how many such paths there are. Because that number can be huge, she is interested in it modulo 998 244 353.

Input

The first line contains two integers n and k ($1 \leq n \leq 500, 1 \leq k \leq 10^6$), which denote the size of the field and Iva's favourite number.

In each of the next n lines, there are n numbers describing the i -th row of the field ($-1 \leq a_{i,j} \leq 10^6$). If $a_{i,j} = -1$, then that cell contains an obstacle, otherwise $1 \leq a_{i,j} \leq 10^6$ and that cell contains a piece of candy with that number.

Output

Print a single line with the required number from the task.

Scoring

Subtask	Points	Constraints
1	13	$n, k, a_{i,j} \leq 20$
2	17	$n, k \leq 20$
3	33	$k \leq 20$
4	47	No additional constraints

Examples

input

2 2
3 2
1 4

output

2

input

3 6
5 2 -1
7 3 6
-1 3 1

output

3

Clarification of the second example:

There are three possible paths such that the product is divisible by 6: $5 \cdot 2 \cdot 3 \cdot 3 \cdot 1$, $5 \cdot 2 \cdot 3 \cdot 6 \cdot 1$, $5 \cdot 7 \cdot 3 \cdot 6 \cdot 1$.



Task Skrivača

Marin and Luka are playing a popular kids game called *Hide and Seek (Skrivača)*. They are playing in their house which has n rooms, and m pairs of rooms are connected by a door. Rooms are labeled from 1 to n and for each pair of rooms there exists a path from one room to another.



Luka has thought of a hiding strategy: when Marin enters room v , Luka will hide in room a_v . At the start of the game Marin chooses his starting room v_0 and Luka hides in the room a_{v_0} . In each step of the game, firstly Marin chooses a room u adjacent to his current room and enters it. After that, Luka knows Marin is in room u so by his hiding strategy he hides in room a_u . Notice that Luka can choose any path to reach room a_u and that in one step of the game he can pass through an arbitrary number of rooms.

Marin will find Luka when both of them are located in the same room, at that moment the game ends.

Marin found out about Luka's hiding strategy so he wants you to determine for each starting room if Marin can find Luka in a finite amount of steps, and if he can, determine the least amount of steps necessary if both of them play optimally. (Marin plays such that he finds Luka in the least amount of steps and Luka plays such that Marin finds him in the most amount of steps).

Input

In the first line there are integers n, m ($1 \leq n \leq 2 \cdot 10^5$, $n - 1 \leq m \leq \min(5 \cdot 10^5, \frac{n \cdot (n-1)}{2})$), the number of rooms and the number of pairs of connected rooms.

In the second line there are n integers a_i ($1 \leq a_i \leq n$), describing Luka's hiding strategy.

In i -th of the next m lines there are integers x_i, y_i ($1 \leq x_i, y_i \leq n$, $x_i \neq y_i$), denoting that room x_i and room y_i are connected. Between each pair of rooms there will be at most one connection.

Output

In the first and only line print n numbers, where the i -th number represents the least amount of steps necessary for Marin to find Luka if Marin starts in room i , or -1 if Marin can't find Luka.

Scoring

Subtask	Points	Constraints
1	15	$n \leq 1\,000$, $m \leq 2\,000$
2	25	$m = n - 1$
3	30	Luka's hiding strategy will be such that he will never attempt to hide in the same or adjacent room to where Marin is currently located, and the structure of the house will be such that the game can end in at most 5 different rooms independent of Luka's hiding strategy
4	40	No additional constraints.



Examples

input

4 4
3 4 1 2
1 2
2 3
3 4
4 1

output

-1 -1 -1 -1

input

8 9
2 3 2 1 6 5 6 7
1 2
1 3
2 4
3 4
4 5
4 6
6 7
5 7
4 8

output

1 2 2 2 1 1 1 1

input

9 8
1 9 1 1 1 9 9 9 1
1 2
2 3
3 4
4 5
5 6
6 7
7 8
8 9

output

0 1 1 2 1 1 2 1 1

Clarification of the second example:

Marin enters room 8 from room 4 in the first step, and in the second he goes back to room 4. Luka needs to pass through room 4 to get from room 7 to room 1 so Marin can find Luka in 2 steps.